Lecture 37: Sections 6.1 and 6.2
Law of Sines and Law of Cosines

Solving oblique triangles using the Law of Sines

Solving oblique triangles using the Law of Cosines

Computing the area of oblique triangles

Heron's Formula

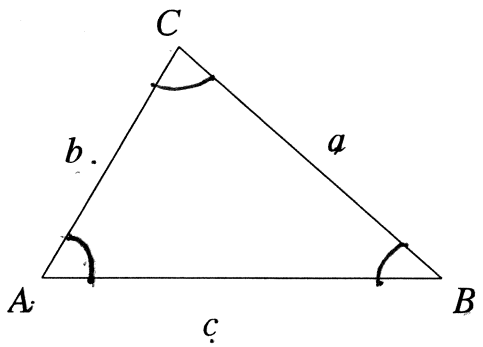
$$\sin 105^\circ = \sin(45^\circ + 60^\circ)$$

$$= \sin 45^\circ \cos 60^\circ + \cos 45^\circ \sin 60^\circ$$

$$= \frac{\sqrt{2}}{2} \cdot \frac{1}{2} + \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{3}}{2}$$

$$\sin 105^\circ = \frac{\sqrt{2}+\sqrt{6}}{4}$$

Def. An **oblique triangle** is a triangle that is not a right triangle.



To Solve an oblique triangle, we have the following four cases.

Law of Sines
1. One side and two angles AAS, ASA
2. Two sides and the angle opposite one of those sides SSA

Law of Cosines
3. Three sides SSS
4. Two sides and the their included angle SAS

Law of Sines

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

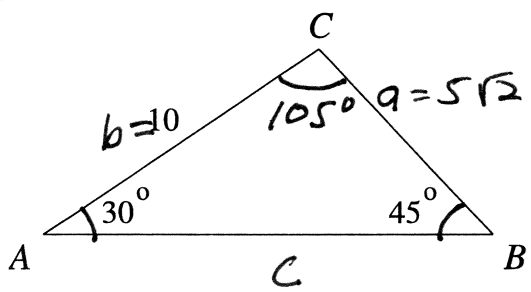
or

reciprocal

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

Two Angles and One Side – AAS

ex. Solve the triangle. $\angle C = 180^\circ - (30^\circ + 45^\circ) = 105^\circ$



$$\frac{a}{\sin A} = \frac{b}{\sin B} \rightarrow \frac{a}{\sin 30^\circ} = \frac{10}{\sin 45^\circ} \rightarrow a = \frac{10 \sin 30^\circ}{\sin 45^\circ} = \frac{\cancel{10}^{5} \cdot \left(\frac{1}{\cancel{2}}\right)}{\frac{\sqrt{2}}{2}}$$

$$= 5 \cdot \frac{2}{\sqrt{2}} = \frac{10}{\sqrt{2}}\left(\frac{\sqrt{2}}{\sqrt{2}}\right) = \frac{\cancel{10}^{5}\sqrt{2}}{\cancel{2}} = \boxed{5\sqrt{2} = a}$$

$$\frac{c}{\sin C} = \frac{b}{\sin B} \rightarrow c = \frac{b \sin C}{\sin B} \rightarrow c = \frac{10 \sin 105^\circ}{\sin 45^\circ} = \frac{\cancel{10}^{5}\left(\frac{\sqrt{6}+\sqrt{2}}{\cancel{2}\,4}\right)}{\frac{\sqrt{2}}{2}}$$

Checkpoint: Lecture 37, problem 1

$$\frac{5(\sqrt{6}+\sqrt{2})}{\cancel{2}} \cdot \frac{\cancel{2}}{\sqrt{2}} = \frac{5(\sqrt{6}+\sqrt{2})}{\sqrt{2}}\left(\frac{\sqrt{2}}{\sqrt{2}}\right) = \frac{5(\sqrt{12}+\sqrt{4})}{2} = \frac{5(2\sqrt{3}+2)}{2}$$

$$= \frac{5 \cdot \cancel{2}(\sqrt{3}+1)}{\cancel{2}} = \boxed{5\sqrt{3}+5}$$

Checkpoint: Lecture 37 - problem 1

In $\triangle ABC$, $\angle A = 34°$, $\angle B = 66°$, and $a = 10$. What does c equal?

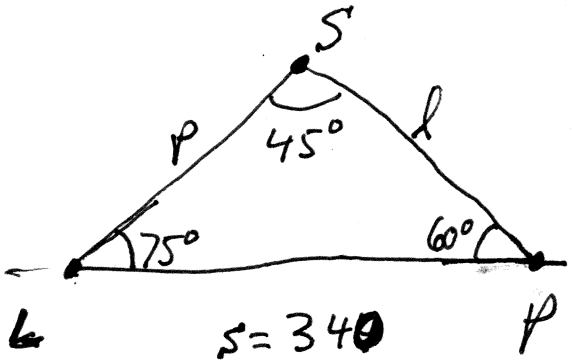
a) $\dfrac{10 \sin 80°}{\sin 34°}$ b) $\dfrac{10 \sin 34°}{\sin 80°}$ c) $\dfrac{10 \sin 80°}{\sin 66°}$

d) $\dfrac{10 \sin 66°}{\sin 80°}$ e) $\dfrac{10 \sin 34°}{\sin 66°}$

$\angle S = 180° - (75° + 60°) = 45°$

Two Angles and One Side – ASA Law of Sines

ex. A satellite orbiting the earth passes directly overhead at the observation stations in Phoenix and Los Angeles, 340 miles apart. At an instant when the satellite is between these two stations, its angle of elevation is simultaneously observed to be 60° at Phoenix and 75° at Los Angeles. How far is the satellite from Los Angeles? *Find p.*



$$\frac{p}{\sin P} = \frac{s}{\sin S} \longrightarrow p = \frac{s \sin P}{\sin S}$$

$$p = \frac{340 \sin 60°}{\sin 45°} = \frac{340\left(\frac{\sqrt{3}}{2}\right)}{\frac{\sqrt{2}}{2}}$$

$$p = 340\left(\frac{\sqrt{3}}{\cancel{2}}\right) \cdot \frac{\cancel{2}}{\sqrt{2}} = \frac{340\sqrt{3}}{\sqrt{2}}\left(\frac{\sqrt{2}}{\sqrt{2}}\right) = \frac{340\sqrt{6}}{2} = \boxed{170\sqrt{6} \text{ mi}}$$

Checkpoint: Lecture 37, problem 2

Checkpoint: Lecture 37 - problem 2

In $\triangle ABC$, $\angle A = 34°$, $\angle B = 66°$, and $c = 10$. What does b equal?

a) $\dfrac{10 \sin 80°}{\sin 34°}$ b) $\dfrac{10 \sin 34°}{\sin 80°}$ c) $\dfrac{10 \sin 80°}{\sin 66°}$

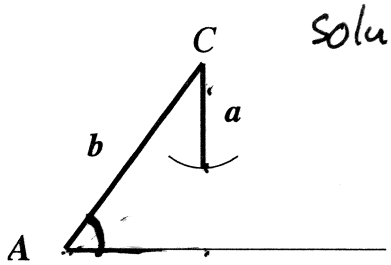
d) $\dfrac{10 \sin 66°}{\sin 80°}$ e) $\dfrac{10 \sin 34°}{\sin 66°}$

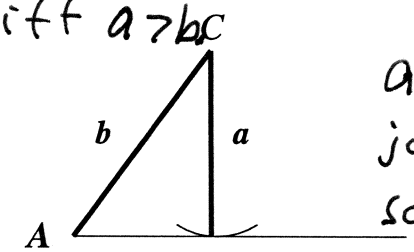
The Ambiguous Case – SSA Law of Sines

Given a, b and A: assuming $\angle A$ is acute.

Note: If $\angle A > 90°$, then there is one solution iff $a > b$

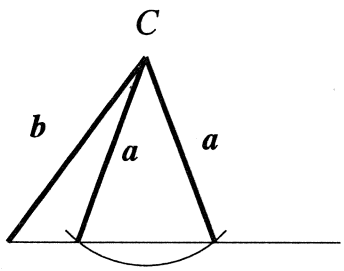
$a < b$, too short, no solution.

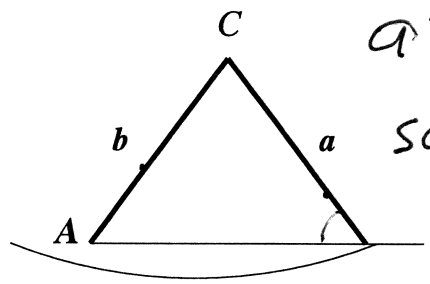




$a < b$, just long enough, so one solution.

$a < b$, a is more than long enough, so two solutions.





$a > b$, so one solution.

ex. Solve triangle ABC, where $\angle A = 45°$, $a = 7\sqrt{2}$ and $b = 7$. Since $a > b$, one solution

Find $\angle B$

$$\frac{\sin B}{b} = \frac{\sin A}{a} \longrightarrow \sin B = \frac{b \sin A}{a} \longrightarrow \sin B = \frac{\cancel{7}\sin 45°}{\cancel{7}\sqrt{2}} = \frac{\frac{\sqrt{2}}{2}}{\frac{\sqrt{2}}{1}}$$

$$\sin B = \frac{\cancel{\sqrt{2}}}{2} \cdot \frac{1}{\cancel{\sqrt{2}}} = \frac{1}{2}$$

thus $\angle B = 30°$

Find $\angle C$: $180° - (45° + 30°) = 105°$

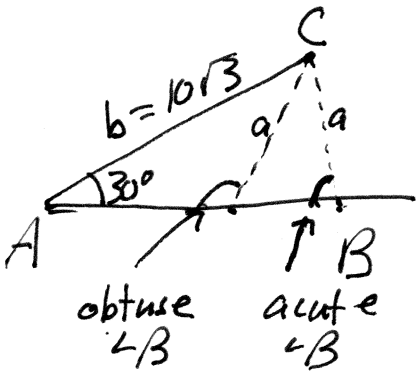
Find side c

$$\frac{c}{\sin C} = \frac{b}{\sin B} \longrightarrow c = \frac{b \sin C}{\sin B} \longrightarrow c = \frac{7 \sin 105°}{\sin 30°} = 7($$

$$c = \frac{7\left(\frac{\sqrt{6}+\sqrt{2}}{4}\right)}{\frac{1}{2}} = 7\left(\frac{\sqrt{6}+\sqrt{2}}{\cancel{4}\,2}\right) \cdot \frac{\cancel{2}}{1} = \boxed{\frac{7(\sqrt{6}+\sqrt{2})}{2} = \frac{7\sqrt{6}+7\sqrt{2}}{2}}$$

NOTE: If $\sin B < 1$, we must check the angle and its supplement as possibilities for $\angle B$, because any angle smaller than $180°$ can be in the triangle.

ex. Solve triangle ABC, where $\angle A = 30°$, $a = 10$ and $b = 10\sqrt{3}$. $\angle A$ is acute, $a < b$, so possibly two solutions



Find $\angle B$: $\dfrac{\sin B}{b} = \dfrac{\sin A}{a} \rightarrow \sin B = \dfrac{b \sin A}{a}$

$\sin B = \dfrac{\cancel{10}\sqrt{3}\sin 30°}{\cancel{10}} = \sqrt{3}\left(\frac{1}{2}\right) = \dfrac{\sqrt{3}}{2}$

so $B = 60°$ or $120°$

Both work since $30° + 60° < 180°$, $30° + 120° < 180°$

solution 1:

$\angle B = 60°$

$\angle C = 180° - (30° + 60°) = \boxed{90°}$

$\dfrac{c}{\sin C} = \dfrac{a}{\sin A} \rightarrow c = \dfrac{a \sin C}{\sin A}$

$c = \dfrac{10 \sin 90°}{\sin 30°} = \dfrac{10 \cdot 1}{\frac{1}{2}} = \boxed{20}$

Solution 2:

$\angle B = 120°$

$\angle C = 180° - (120° + 30°) = \boxed{30°}$

Find c.

$c = \dfrac{10 \cancel{\sin 30°}}{\cancel{\sin 30°}} = \boxed{10}$

Checkpoint: Lecture 37 - problem 3

In $\triangle ABC$, $\angle A = 30°$, $a = 6$, and $b = 6\sqrt{3}$. What does c equal?

a) $c = 6$ only b) $c = 12$ only

c) $c = 6$ or $c = 12$ (cannot determine which)

d) There is no value for c that works

Three Sides – SSS Law of Cosines

ex. Find all angles of triangle ABC with $a = 4$, $b = 5$, and $c = 8$.

Find $\angle A$.

$a^2 = b^2 + c^2 - 2bc\cos A$

$4^2 = 5^2 + 8^2 - 2(5)(8)\cos A$

$16 = 25 + 64 - 80\cos A$

$-73 = -80\cos A$

$\cos A = \frac{73}{80}$

$\angle A = \cos^{-1}\left(\frac{73}{80}\right) \overset{\text{calc.}}{\approx} 24.15°$

Find $\angle B$

$b^2 = a^2 + c^2 - 2ac\cos B$

$25 = 16 + 64 - 2(4)8\cos B$

$-55 = -64\cos B$

$\cos B = \frac{55}{64}$

$\boxed{\angle B = \cos^{-1}\left(\frac{55}{64}\right) \overset{\text{calc.}}{\approx} 30.75°}$

Find $\angle C$

$c^2 = a^2 + b^2 - 2ab\cos C$

$64 = 16 + 25 - 2(4)5\cos C$

$23 = -40\cos C$

$\cos C = -\frac{23}{40}$

$\angle C = \cos^{-1}\left(-\frac{23}{40}\right) \overset{\text{calc.}}{\approx} 125.10°$

Two Sides and the Included Angle – SAS

ex. Solve triangle ABC, where $\angle A = 45°$, $b = 18$ and $c = 10$. First use Law of Cosines to find a.

$a^2 = b^2 + c^2 - 2bc\cos A$

$a^2 = 18^2 + 10^2 - 2(18)(10)\cos 45°$

$a^2 = 324 + 100 - 360\frac{\sqrt{2}}{2}$

$a^2 = 424 - \overset{180}{\cancel{360}}\frac{\sqrt{2}}{\cancel{2}}$

$a^2 = 424 - 180\sqrt{2}$

$\boxed{a = \sqrt{424 - 180\sqrt{2}}} \approx 13.02$

To finish use the Law of Cosines to find $\angle B$ and $\angle C$.

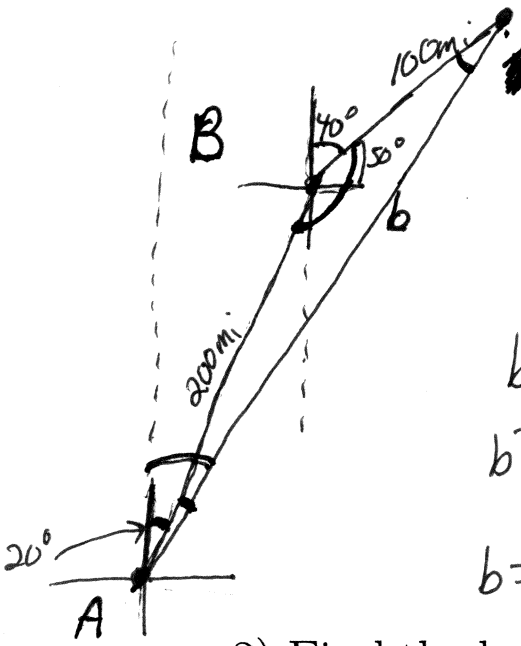
Do this on your own.

$\angle B = \cos^{-1}\left(\frac{14 - 45\sqrt{2}}{5\sqrt{424 - 180\sqrt{2}}}\right) \overset{\text{calc.}}{\approx} 102.10°$

$\angle C = \cos^{-1}\left(\frac{14 - 5\sqrt{2}}{\sqrt{424 - 180\sqrt{2}}}\right) \overset{\text{calc.}}{\approx} 32.90°$

ex. A pilot sets out from an airport and heads in the direction N20°E, flying at 200 mi/h. After one hour, he makes a course correction and heads in the direction N40°E. Half an hour after that, engine trouble forces him to make an emergency landing.

1) Find the distance between the airport and his final landing point. $\angle B = 50^\circ + 90^\circ + 20 = 160^\circ$ SAS



want side b

$$b^2 = a^2 + c^2 - 2ac\cos B$$

$$b^2 = 100^2 + 200^2 - 2(100)(200)\cos 160^\circ$$

$$b^2 = 10{,}000 + 40{,}000 - 40{,}000\cos 160^\circ$$

$$b^2 = 50{,}000 - 40{,}000\cos 160^\circ = 10{,}000(5 + 4\cos 160^\circ)$$

$$b = \sqrt{100^2(5+4\cos 160^\circ)} = \boxed{100\sqrt{5+4\cos 160^\circ}} \overset{\text{calc}}{\approx} 295.95 \text{ mi}$$

2) Find the bearing from the airport to his final landing point. Answer is $N(20^\circ + \angle A)\,E. = \boxed{N\ 28^\circ\ E}$

$$a^2 = b^2 + c^2 - 2bc\cos A$$

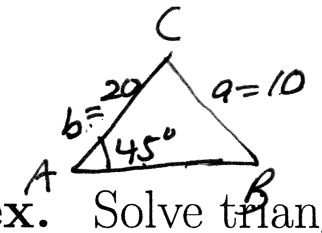
$$100^2 = \left(100\sqrt{5+4\cos 160^\circ}\right)^2 + 200^2 - 100\sqrt{5+4\cos 160^\circ}\,(200)\cos A$$

$$\left(100^2 = 100^2(5+4\cos 160^\circ) + 200^2 - 20{,}000\sqrt{5+4\cos 160^\circ}\cos A\right)\frac{1}{100^2}$$

Checkpoint: Lecture 37, problem 4

$$1 = 5 - 4\cos 160^\circ + 4 - 4\sqrt{5-4\cos 160^\circ}\cos A$$

$$\cos A = \frac{8 - 4\cos 160^\circ}{4\sqrt{5-4\cos 160^\circ}} \rightarrow \angle A = \cos^{-1}\left(\frac{2-\cos 160^\circ}{\sqrt{5-4\cos 160^\circ}}\right) \overset{\text{calc}}{\approx} 6.636^\circ \approx 7$$



ex. Solve triangle ABC, where $\angle A = 45°$, $a = 10$ and $b = 20$.

Find $\angle B$

$$\frac{\sin B}{b} = \frac{\sin A}{a} \rightarrow \sin B = \frac{b \sin A}{a} \rightarrow \sin B = \frac{\overset{2}{\cancel{20}} \sin 45°}{\cancel{10}}$$

$$\sin B = \cancel{2}\frac{\sqrt{2}}{\cancel{2}} = \sqrt{2} \qquad \sin B = \sqrt{2}$$

$-1 \le \sin B \le 1$. $\sqrt{2} > 1$ So no solution

No triangle exists with the given dimensions.

Checkpoint: Lecture 37, problem 3

Law of Cosines SSS, SAS

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$b^2 = a^2 + c^2 - 2ac \cos B$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

NOTE: If $\angle C$ is a right angle, then $\cos C = 0$ and the Law of Cosines reduces to the Pythagorean Theorem, $c^2 = a^2 + b^2$.

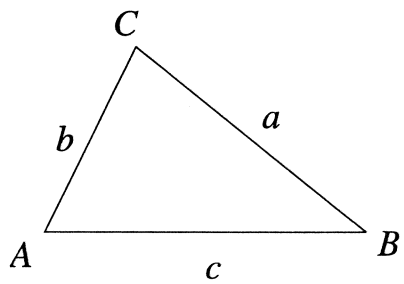
Checkpoint: Lecture 37 - problem 4

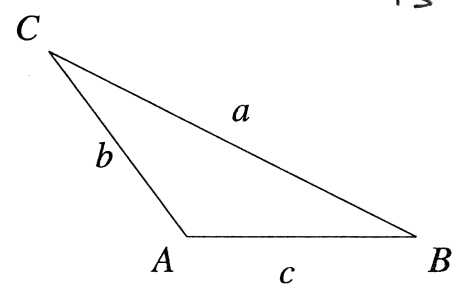
In $\triangle ABC$, $a = 5$, $b = 6$, and $c = 7$. What is $\cos B$?

a) $-\frac{19}{35}$ b) $-\frac{3}{35}$ c) $\frac{3}{35}$ d) $\frac{19}{35}$ e) $\frac{38}{35}$

Area of an Oblique Triangle

$$\text{Area} = \frac{1}{2}bc\sin A = \frac{1}{2}ab\sin C = \frac{1}{2}ac\sin B$$ if SAS is known.





Heron's Area Formula

The area of triangle ABC is given by if SSS is known

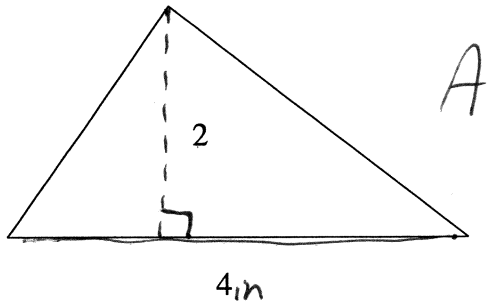
$$\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}$$

where $s = \frac{1}{2}(a+b+c)$ is the **semiperimeter** of the triangle.

half

ex. Find the area of the triangle:

1) $A = \frac{1}{2}bh = \frac{1}{2}$ base · height

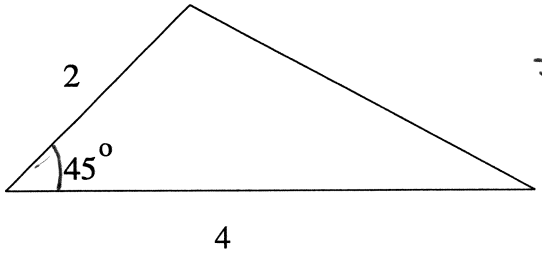


$A = \frac{1}{2}(4)(2) = \boxed{4 \text{ sq inches}}$

2) SAS

$\text{Area} = \frac{1}{2}bc\sin A$

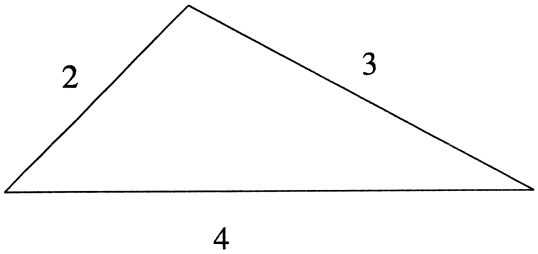
$= \frac{1}{2}\cdot 2\cdot 4\cdot \sin 45^\circ$



$= \frac{\cancel{4}^{2}\sqrt{2}}{\cancel{2}} = \boxed{2\sqrt{2} \text{ sq in}}$

3) SSS

$s = \frac{1}{2}(2+3+4) = \frac{9}{2}$



$\text{Area} = \sqrt{s(s-2)(s-3)(s-4)}$

$A = \sqrt{\frac{9}{2}\left(\frac{9}{2}-2\right)\left(\frac{9}{2}-3\right)\left(\frac{9}{2}-4\right)}$

$A = \sqrt{\frac{9}{2}\cdot\frac{5}{2}\cdot\frac{3}{2}\cdot\frac{1}{2}} = \boxed{\frac{3\sqrt{15}}{4} \text{ sq in.}}$

Checkpoint: Lecture 37, problem 5

Checkpoint: Lecture 37 - problem 5

In $\triangle ABC$, $\angle A = 60°$, $b = 5$, and $c = 12$.
In $\triangle DEF$, $d = 7$, $e = 10$, and $f = 13$.
What is the sum of the areas of $\triangle ABC$ and $\triangle DEF$?

a) $25\sqrt{3}$ b) $30\sqrt{3}$ c) $35\sqrt{3}$

d) $50\sqrt{3}$ e) $60\sqrt{3}$